Lateral-Torsional Response of Unbraced Single-Span Z-Purlins

Z8x2 16 ga (0.5 in lips) and Z8x2.5 16 ga (0.75 in lips)
25 ft simple span, 50 plf gravity load on the top flange

Analysis with PlautBeamElement.jl v0.4.0 · November 06, 2026

1 Introduction

This study computes the deflections, twist and internal forces of two unbraced cold-formed steel Z-purlins (Z8x2 and Z8x2.5, 16 ga) under a 50 plf gravity load on the top flange, and checks whether that load is below the lateral-torsional buckling load. Because $I_{xy} \neq 0$ and the load is eccentric to the shear center, a vertical load produces lateral deflection and twist as well as vertical deflection. The finite element (FE) formulation of Plaut and Moen [1] captures these coupled effects.

2 Problem Statement

2.1 Purlins and geometry

Dimensions are from the RunToSolve purlin databases (Table 1). Centerline dimensions are $h = H - t$ and $b = B - t$; corner radii are ignored.

Table 1: Purlin geometry.

	Z8x2 16 ga, 0.5 in lips	**Z8x2.5 16 ga, 0.75 in lips**
Database designation	Z8x2 060 0.5lip	Z8x2.5 060 0.75 lip
Database source	PurlinLine.jl database/Purlins.csv	RoofHugger_PlautBeamFE.jl database/Purlins.csv
Thickness t (in)	0.06	0.06
Depth H, out-to-out (in)	8	8
Flange width B, out-to-out (in)	2	2.5
Lip length D (in)	0.5	0.75
Lip angle to flange	55°	55°
Centerline web depth h (in)	7.94	7.94
Centerline flange width b (in)	1.94	2.44

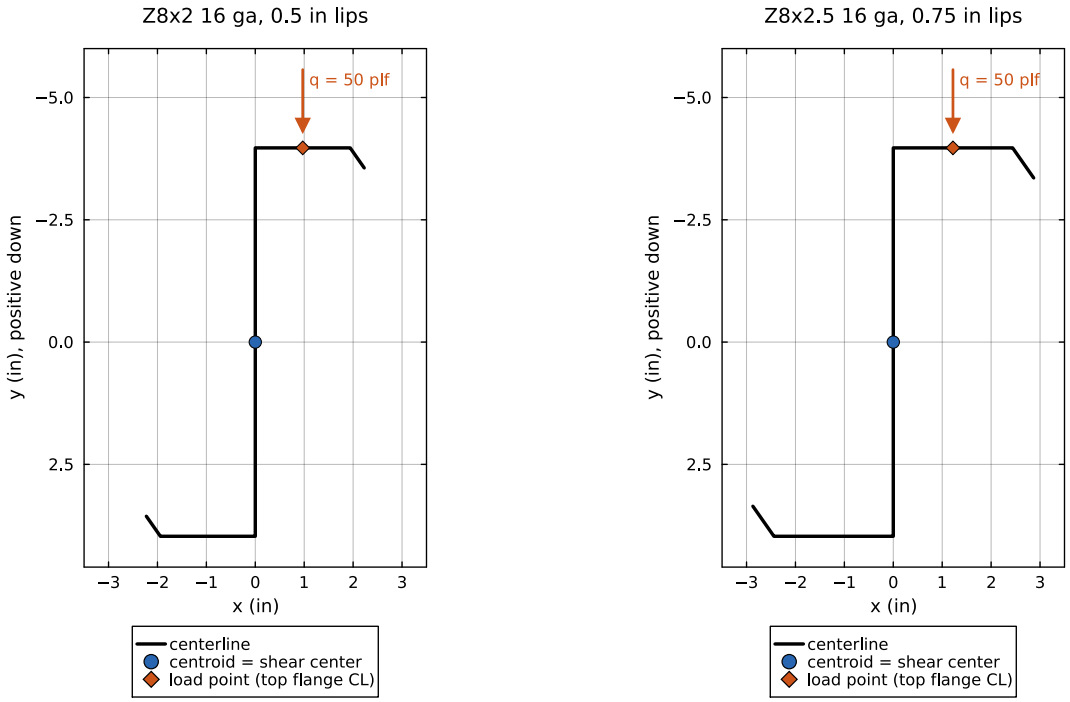


Figure 1: Centerline models. x points toward the top flange tip; y is positive downward.

2.2 Span, supports and loading

Table 2: Analysis inputs (kip, in).

Input	Value
Span	$L = 25$ ft $= 300$ in, single simple span
Load	$q = 50$ plf $= 0.004167$ kip/in, gravity, uniform over the full span
Load location	Centerline of the top flange (a_x, a_y from the shear center, Table 3)
Roof slope	$\theta = 0$ (load acts parallel to the web), so $q_x = 0$ and $q_y = q$
Supports at $z = 0$ and $z = L$	Pinned: $u = v = \varphi = 0$; u', v' and warping φ' free
Axial support	Axial translation restrained at a single node at midspan
Continuous bracing	None: $k_x = 0$, $k_\varphi = 0$
Material	$E = 29500$ ksi, $G = 11300$ ksi
Mesh	50 equal two-node elements, $L_e = 6$ in

End supports use the DOF tuple `("fixed","free","fixed","free","fixed","free")` for $(u, u', v, v', \varphi, \varphi')$. The midspan axial restraint is not modeled (Section 3.7).

2.3 Section properties

Computed as in Section 3.1 (worked in Appendix A.1). For a Z-section the shear center coincides with the centroid.

Table 3: Section properties about the centroidal x–y axes (centerline model, sharp corners).

Property	Units	Z8x2 16 ga, 0.5 in lips	Z8x2.5 16 ga, 0.75 in lips
Area A	in^2	0.7692	0.8592
I_x	in^4	7.023	8.328
I_y	in^4	0.5529	1.217
I_{xy}	in^4	−1.367	−2.291
J	in^4	0.000923	0.001031
C_w	in^6	6.521	13.81
Shear center (x_s, y_s)	in	(0, 0)	(0, 0)
a_x (load, $+x$ of shear center)	in	0.97	1.22
a_y (load, above shear center)	in	3.97	3.97

3 Analysis

3.1 Section properties

Each section is five straight centerline segments of thickness t. For linear f, g over a segment of length ℓ from node i to j:

$$\int_{\text{seg}} fg \,\mathrm{d}A = \frac{t\ell}{6}\big(2f_ig_i + f_ig_j + f_jg_i + 2f_jg_j\big). \tag{1}$$

This gives I_x, I_y, I_{xy}; $J = \sum \ell t^3/3$; and $C_w = \int \omega_n^2 \,\mathrm{d}A$, with ω_n the normalized sectorial coordinate about the shear center.

3.2 Governing equations

Plaut and Moen [1]:

$$\begin{aligned} EI_y u'''' + EI_{xy} v'''' + k_x\big(u + a_y\varphi\big) &= q_x \\ EI_x v'''' + EI_{xy} u'''' &= q_y \\ EC_w\varphi'''' - GJ\varphi'' + k_x a_y\big(u + a_y\varphi\big) + k_\varphi\varphi &= q_x\big(a_y - a_x\varphi\big) + q_y\big(a_x + a_y\varphi\big) \end{aligned} \tag{2}$$

With $q_x = 0$, $q_y = q$ and $k_x = k_\varphi = 0$:

$$EI_y u'''' + EI_{xy} v'''' = 0, \quad EI_x v'''' + EI_{xy} u'''' = q, \quad EC_w\varphi'''' - GJ\varphi'' - qa_y\varphi = qa_x. \tag{3}$$

Bending is coupled through I_{xy}; torsion is uncoupled from bending since $k_x = 0$. qa_x is the eccentric torque and $-qa_y\varphi$ is the load-height (second-order) effect.

3.3 Finite element discretization

50 two-node elements with cubic Hermite interpolation of u, v, φ; nodal DOFs $(u, u', v, v', \varphi, \varphi')$, with $\xi = (z - z_1)/L_e$:

$$\boldsymbol{N} = \big[1 - 3\xi^2 + 2\xi^3, \quad L_e(\xi - 2\xi^2 + \xi^3), \quad 3\xi^2 - 2\xi^3, \quad L_e(\xi^3 - \xi^2)\big]. \tag{4}$$

Stiffness kernels: $\int \boldsymbol{N}''^T\boldsymbol{N}''$ (EI, EC_w) and $\int \boldsymbol{N}'^T\boldsymbol{N}'$ (GJ). The load-height term gives the geometric stiffness

$$\boldsymbol{k}_g = \int_0^{L_e} (q_x a_x - q_y a_y)\boldsymbol{N}^T\boldsymbol{N}\,\mathrm{d}z = -qa_y\int_0^{L_e} \boldsymbol{N}^T\boldsymbol{N}\,\mathrm{d}z, \tag{5}$$

(softening for gravity load). Loads: $\int q\boldsymbol{N}^T$ on v and $\int qa_x\boldsymbol{N}^T$ on φ (Appendix A.2).

3.4 Solution: first- and second-order analyses

$$\big[\boldsymbol{K}_e + \gamma\boldsymbol{K}_g(q)\big]\boldsymbol{d} = \boldsymbol{F}(q), \tag{6}$$

solved by sparse LU factorization. First order: $\gamma = 0$. Second order: $\gamma = 1$, valid only below the critical load.

3.5 Stability check

q_{cr} is the smallest μ making $\boldsymbol{K}_e + \mu\boldsymbol{K}_g(q=1)$ singular. Closed form ($\lambda_1 = \pi/L$):

$$q_{\text{cr}} = \frac{EC_w\lambda_1^4 + GJ\lambda_1^2}{a_y}. \tag{7}$$

3.6 Internal forces

Recovered at element ends by equilibrium:

$$\boldsymbol{g} = \big[\boldsymbol{k}_e + \gamma\boldsymbol{k}_g\big]\boldsymbol{d}_e - \boldsymbol{f}_e, \quad \text{end 1:} -\boldsymbol{g}_{1..6}, \quad \text{end 2:} +\boldsymbol{g}_{7..12}. \tag{8}$$

giving $V_x,\ M_y = E\big(I_y u'' + I_{xy} v''\big),\ V_y,\ M_x = E\big(I_x v'' + I_{xy} u''\big),\ T = GJ\varphi' - EC_w\varphi'''$ and $B = EC_w\varphi''$. Sagging M_x is negative (y down).

3.7 Axial restraint at midspan

The formulation has no axial DOF. With no axial load, the restraint only prevents rigid-body sliding and does not affect any result.

4 Results

4.1 Stability

Table 4: Lateral-torsional stability of the unbraced purlins.

	Z8x2 16 ga, 0.5 in lips	**Z8x2.5 16 ga, 0.75 in lips**
Applied load q (plf)	50	50
q_{cr}, closed form Equation 7 (plf)	10.45	18.67
q_{cr}, FE eigenvalue (plf)	10.45	18.67
q/q_{cr}	4.78	2.68
$\boldsymbol{K}_e + \boldsymbol{K}_g$ positive definite at q?	**No**	**No**

Unbraced, neither purlin can carry 50 plf; both buckle first. The Z8x2.5′s larger C_w raises q_{cr} by 79 %.

4.2 Deformations

4.2.1 Z8x2 16 ga (0.5 in lips)

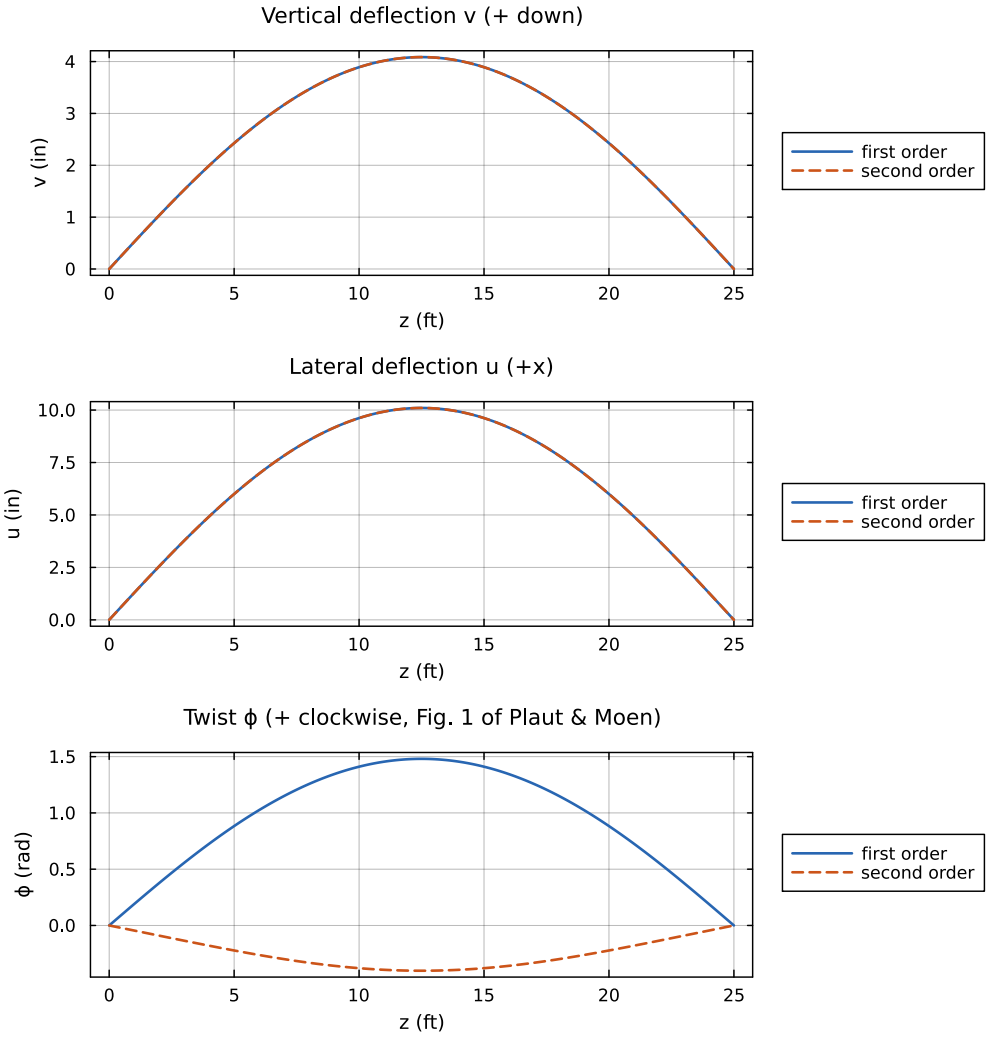


Figure 2: Z8x2 16 ga (0.5 in lips): vertical deflection, lateral deflection and twist along the span.

4.2.2 Z8x2.5 16 ga (0.75 in lips)

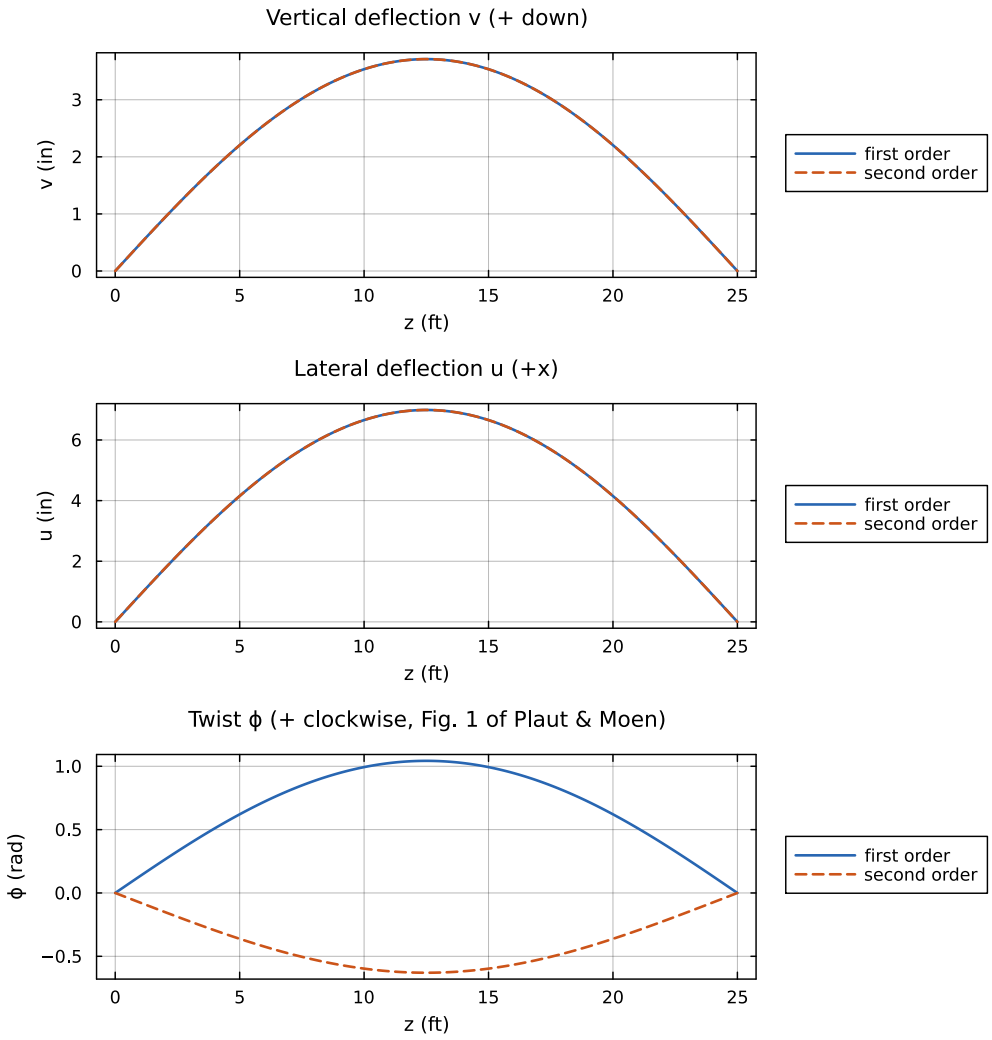


Figure 3: Z8x2.5 16 ga (0.75 in lips): vertical deflection, lateral deflection and twist along the span.

4.2.3 Discussion of deformations

Table 5: Midspan deformations.

	Z8x2 16 ga, 0.5 in lips		**Z8x2.5 16 ga, 0.75 in lips**	
Midspan	1st order	2nd order	1st order	2nd order
v (in)	4.086	4.086	3.712	3.712
u (in)	10.1	10.1	6.989	6.989
φ (rad)	1.48	−0.4022	1.043	−0.6296
L/v	73	73	81	81

v and u are the same in both analyses and match the closed form (Appendix A.3). $u/v = -I_{xy}/I_y$ (2.47 and 1.88). The Z8x2.5 reduces u by 31 % and v by 9 %.

The twist is not reliable: first-order values exceed 1 rad, and the second-order twist reverses sign because $q > q_{cr}$ (Appendix A.3).

4.3 Internal forces

4.3.1 Z8x2 16 ga (0.5 in lips)

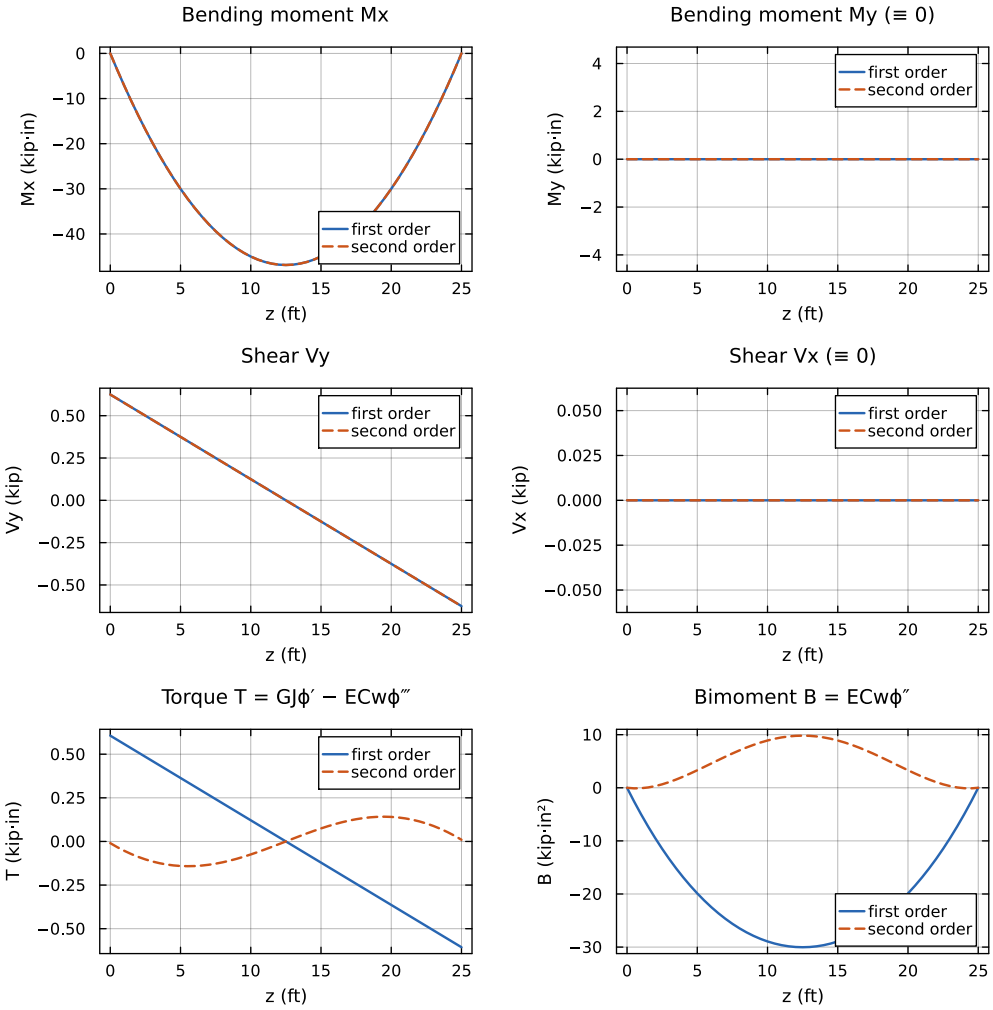


Figure 4: Z8x2 16 ga (0.5 in lips): internal forces along the span.

4.3.2 Z8x2.5 16 ga (0.75 in lips)

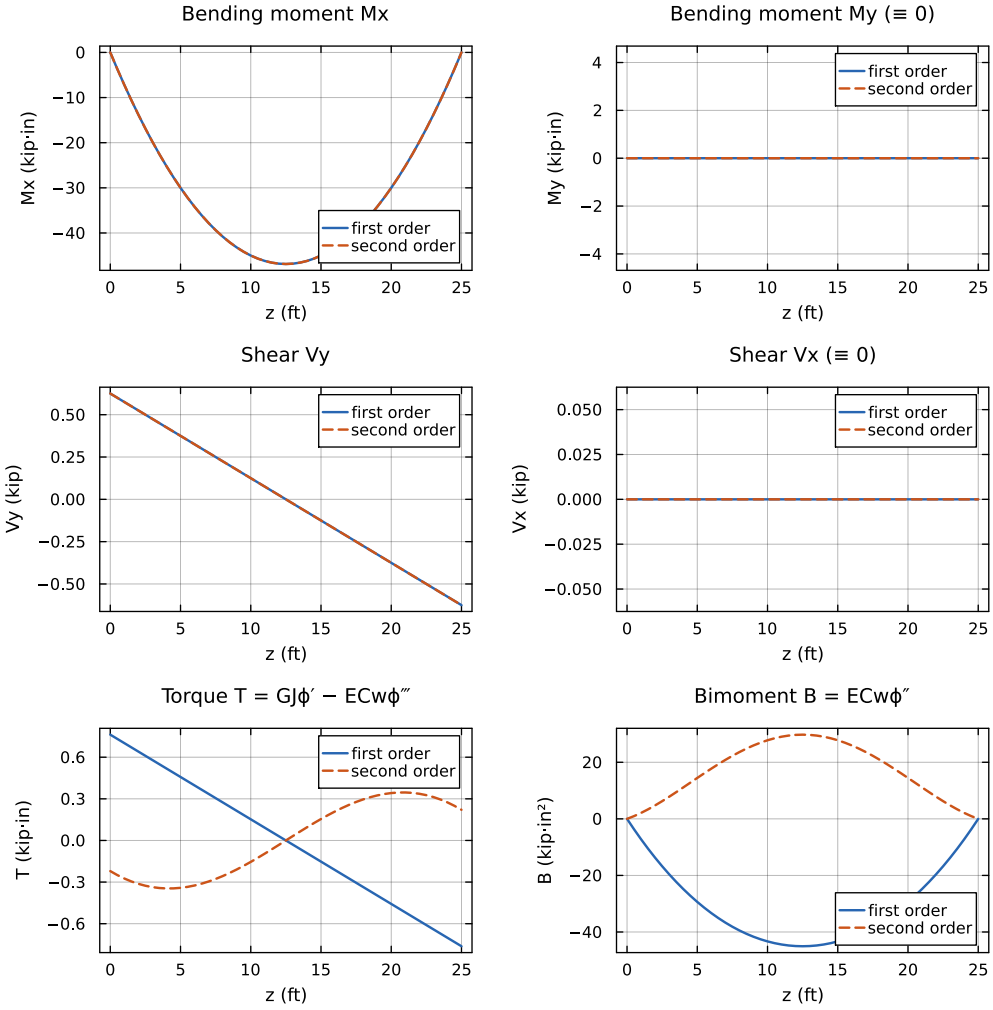


Figure 5: Z8x2.5 16 ga (0.75 in lips): internal forces along the span.

4.3.3 Discussion of internal forces

Table 6: Key internal forces. $M_x < 0$ is sagging in the y-down convention.

	Z8x2 16 ga, 0.5 in lips		**Z8x2.5 16 ga, 0.75 in lips**	
Quantity	1st order	2nd order	1st order	2nd order
$M_x(L/2)$ (kip·in)	−46.875	−46.875	−46.875	−46.875
$V_y(0)$ (kip)	0.625	0.625	0.625	0.625
$T(0)$ (kip·in)	0.6062	−0.009756	0.7625	−0.2209
$B(L/2)$ (kip·in²)	−30.03	9.817	−45.04	29.76
max $\lvert M_y \rvert$ (kip·in)	≈ 0	≈ 0	≈ 0	≈ 0
max $\lvert V_x \rvert$ (kip)	≈ 0	≈ 0	≈ 0	≈ 0

M_x and V_y are the simple-beam values $(-qL^2/8, qL/2)$ in all cases. $M_y = V_x = 0$ because $q_x = 0$. First-order $T(0) = qa_xL/2$, and B peaks at midspan. Second-order T and B are post-critical and not meaningful.

4.4 Comparison and conclusions

- Unbraced, neither purlin carries 50 plf: q_{cr} = 10.45 plf (Z8x2), 18.67 plf (Z8x2.5).

- v, u, M_x and V_y are valid. The Z8x2.5 is stiffer (v = 3.71 vs 4.09 in; u = 6.99 vs 10.1 in).
- Twist, torque and bimoment are not valid magnitudes at this load.
- Deck bracing (k_x, k_φ) must be included for realistic torsional results.

References

[1] Plaut, Raymond H., Moen, and Cristopher D., "Lateral-torsional deformations of single-span and two-span thin-walled beams with continuous bracing," *Journal of Constructional Steel Research*, vol. 179, p. 106534, 2021, doi: 10.1016/j.jcsr.2021.106534.

A Sample Computations

Worked for the Z8x2; checks given for both sections. Units: kip, in.

A.1 Section properties

Nodes (Table 8): $h = 7.94$ in, $b = 1.94$ in; lip tips at $(\pm(b + D\cos 55°), \pm(h/2 - D\sin 55°))$.

Area and centroid. $A = t\sum \ell = 0.06 \times 12.82 = 0.7692$ in². $x_c = y_c = 0$.

Moments of inertia (Table 7):

- Web ($x = 0$, y from $h/2$ to $-h/2$): $I_x = th^3/12 = 0.06 \times 7.94^3/12 = 2.5028$ in⁴.
- Top flange ($y = -h/2$, x from 0 to b): $I_x = tb(h/2)^2 = 1.8346$ in⁴, $I_y = tb^3/3 = 0.146$ in⁴, $I_{xy} = t(b^2/2)(-h/2) = -0.4482$ in⁴.

Torsion constant. $J = t^3/3\sum \ell = 0.06^3/3 \times 12.82 = 0.000923$ in⁴.

Table 7: Z8x2: segment contributions about the centroid.

Segment	ℓ (in)	A (in²)	I_x (in⁴)	I_y (in⁴)	I_{xy} (in⁴)	J (in⁴)
1 bottom lip	0.5	0.03	0.4257	0.1304	−0.235	0.000036
2 bottom flange	1.94	0.1164	1.8346	0.146	−0.4482	0.0001397
3 web	7.94	0.4764	2.5028	0	0	0.0005717
4 top flange	1.94	0.1164	1.8346	0.146	−0.4482	0.0001397
5 top lip	0.5	0.03	0.4257	0.1304	−0.235	0.000036
Total	12.82	**0.7692**	**7.0234**	**0.5529**	**−1.3666**	**0.000923**

Warping constant. Each segment adds $\Delta\omega_C = X_iY_j - X_jY_i$. $I_{\omega x} = I_{\omega y} \approx 0$, so the shear center is at the centroid. Subtracting the mean gives ω_n (Table 8):

$$C_w = \int \omega_n^2 \, \mathrm{d}A = \sum_{\text{seg}} \frac{t\ell}{3}\left(\omega_i^2 + \omega_i\omega_j + \omega_j^2\right) = 6.521 \text{ in}^6. \tag{9}$$

Table 8: Z8x2: centerline nodes and sectorial coordinates.

Node	x (in)	y (in)	$X = x - x_c$	ω_C (in²)	ω_n (in²)
1	−2.2268	3.5604	−2.2268	0	7.7933
2	−1.94	3.97	−1.94	−1.9331	5.8602
3	0	3.97	0	−9.6349	−1.8416
4	0	−3.97	0	−9.6349	−1.8416
5	1.94	−3.97	1.94	−1.9331	5.8602
6	2.2268	−3.5604	2.2268	0	7.7933

Load location. $a_x = b/2 = 0.97$ in, $a_y = h/2 = 3.97$ in. Z8x2.5: Table 9, Table 10.

Table 9: Z8x2.5: segment contributions about the centroid.

Segment	ℓ (in)	A (in²)	I_x (in⁴)	I_y (in⁴)	I_{xy} (in⁴)	J (in⁴)
1 bottom lip	0.75	0.045	0.6051	0.3179	−0.4366	0.000054
2 bottom flange	2.44	0.1464	2.3074	0.2905	−0.7091	0.0001757
3 web	7.94	0.4764	2.5028	0	0	0.0005717
4 top flange	2.44	0.1464	2.3074	0.2905	−0.7091	0.0001757
5 top lip	0.75	0.045	0.6051	0.3179	−0.4366	0.000054
Total	14.32	**0.8592**	**8.3279**	**1.2169**	**−2.2914**	**0.001031**

Table 10: Z8x2.5: centerline nodes and sectorial coordinates.

Node	x (in)	y (in)	$X = x - x_c$	ω_C (in²)	ω_n (in²)
1	−2.8702	3.3556	−2.8702	0	10.0605
2	−2.44	3.97	−2.44	−3.2069	6.8536
3	0	3.97	0	−12.8937	−2.8332
4	0	−3.97	0	−12.8937	−2.8332
5	2.44	−3.97	2.44	−3.2069	6.8536
6	2.8702	−3.3556	2.8702	0	10.0605

A.2 Element matrices and load vector

$L_e = 300/50 = 6$ in. Z8x2, v–v block with $EI_x = 207191$ kip·in²:

$$\boldsymbol{k}_{vv} = \frac{EI_x}{L_e^3}\begin{pmatrix} 12 & 6L_e & -12 & 6L_e \\ 6L_e & 4L_e^2 & -6L_e & 2L_e^2 \\ -12 & -6L_e & 12 & -6L_e \\ 6L_e & 2L_e^2 & -6L_e & 4L_e^2 \end{pmatrix}, \quad \frac{12EI_x}{L_e^3} = 11511 \text{ kip/in.} \tag{10}$$

Other blocks use EI_y, EI_{xy}, EC_w; GJ adds $GJ/(30L_e)[36, 3L_e, -36, 3L_e; \ldots]$. Consistent loads:

$$\boldsymbol{f}_v = q\left[\frac{L_e}{2}, \frac{L_e^2}{12}, \frac{L_e}{2}, -\frac{L_e^2}{12}\right] = [0.0125,\ 0.0125,\ 0.0125,\ -0.0125], \tag{11}$$

$$\boldsymbol{f}_\varphi = qa_x\left[\frac{L_e}{2}, \frac{L_e^2}{12}, \frac{L_e}{2}, -\frac{L_e^2}{12}\right] = [0.01213,\ 0.01213,\ 0.01213,\ -0.01213]. \tag{12}$$

Geometric stiffness: $\boldsymbol{k}_g = -qa_y(L_e/420)[156, 22L_e, 54, -13L_e; \ldots]$, leading term −0.03686 kip·in.

A.3 Deformations: closed-form check

Bending ($\Delta = I_xI_y - I_{xy}^2$):

$$v\left(\frac{L}{2}\right) = \frac{5qL^4}{384E}\frac{I_y}{\Delta}, \quad u\left(\frac{L}{2}\right) = -\frac{5qL^4}{384E}\frac{I_{xy}}{\Delta}. \tag{13}$$

Z8x2: $\Delta = 7.0234 \times 0.5529 - (-1.3666)^2 = 2.0157$ in⁸ and $5qL^4/(384E) = 14.9$ in⁵, so $v(L/2) = 4.086$ in and $u(L/2) = 10.1$ in.

Twist. $\varphi(z) = \sum_{n\ \text{odd}} \varphi_n \sin(\lambda_n z)$, $\lambda_n = n\pi/L$:

$$\varphi_n = \frac{4qa_x/(n\pi)}{k_n - \gamma q a_y}, \quad k_n = EC_w\lambda_n^4 + GJ\lambda_n^2, \tag{14}$$

Z8x2: $qa_y = 0.01654$ kip (Table 11).

Table 11: Z8x2: sine-series terms of the midspan twist.

n	k_n (kip)	$k_n - qa_y$ (kip)	$\varphi_n \sin(n\pi/2)$, 1st	$\varphi_n \sin(n\pi/2)$, 2nd
1	0.003457	−0.01308	1.488	−0.3933
3	0.1977	0.1811	−0.008677	−0.009469
5	1.475	1.458	0.000698	0.0007059
7	5.611	5.594	−0.000131	−0.0001314
9	15.27	15.26	0.00003744	0.00003748
Series sum			1.48	−0.4022
FE			1.48	−0.4022

Since $q > q_{\text{cr}} = k_1/a_y$, $k_1 - qa_y < 0$ and the second-order twist reverses sign.

Table 12: Deformations: closed form vs. FE.

	Z8x2 16 ga, 0.5 in lips		**Z8x2.5 16 ga, 0.75 in lips**	
Midspan	Closed form	FE	Closed form	FE
v (in)	4.08605	4.08605	3.71192	3.71192
u (in)	10.0993	10.0993	6.98947	6.98947
φ 1st order (rad)	1.4804	1.4804	1.0428	1.0428
φ 2nd order (rad)	−0.40216	−0.40216	−0.62962	−0.62962

A.4 Internal forces: element-end recovery

Equation 8 for Z8x2 element 1, first order:

Table 13: Z8x2, element 1, first order: recovery of the end forces (kip, in).

DOF	d_e	$(k_e d_e)$	f_e	$g = kd - f$	**End 1 force** $= -g$
u_1 (in)	≈ 0	≈ 0	≈ 0	≈ 0	$V_x \approx 0$
u_1' (rad)	0.1077	≈ 0	≈ 0	≈ 0	$M_y \approx 0$
v_1 (in)	≈ 0	−0.6125	0.0125	−0.625	$V_y = 0.625$
v_1' (rad)	0.04358	0.0125	0.0125	≈ 0	$M_x \approx 0$
φ_1 (rad)	≈ 0	−0.5941	0.01213	−0.6062	$T = 0.6062$
φ_1' (rad/in)	0.01591	0.01213	0.01213	≈ 0	$B \approx 0$
u_2 (in)	0.6458	≈ 0	≈ 0	≈ 0	
u_2' (rad)	0.1075	≈ 0	≈ 0	≈ 0	
v_2 (in)	0.2613	0.6125	0.0125	0.6	
v_2' (rad)	0.04348	−3.687	−0.0125	−3.675	
φ_2 (rad)	0.0954	0.5941	0.01213	0.582	
φ_2' (rad/in)	0.01587	−2.582	−0.01213	−2.57	

Statics: $V_y(0) = qL/2 = 0.625$ kip; $T(0) = qa_x L/2 = 0.6063$ kip· in; $M_x(0) = B(0) = 0$; $M_x(L/2) = -qL^2/8 = -46.875$ kip·in. Torque and bimoment series:

$$T(z) = \sum_{n\ \text{odd}} \varphi_n \lambda_n (GJ + EC_w \lambda_n^2) \cos(\lambda_n z), \quad B(z) = -EC_w \sum_{n\ \text{odd}} \varphi_n \lambda_n^2 \sin(\lambda_n z). \tag{15}$$

Table 14: Internal forces: closed form vs. FE.

	Z8x2 16 ga, 0.5 in lips		**Z8x2.5 16 ga, 0.75 in lips**	
Quantity	Closed form	FE	Closed form	FE
$M_x(L/2)$ (kip·in)	−46.875	−46.875	−46.875	−46.875
$V_y(0)$ (kip)	0.625	0.625	0.625	0.625
$T(0)$, 1st order (kip·in)	0.60624	0.60625	0.76248	0.7625
$T(0)$, 2nd order (kip·in)	−0.00977	−0.00976	−0.221	−0.221
$B(L/2)$, 1st order (kip·in²)	−30.028	−30.028	−45.038	−45.038
$B(L/2)$, 2nd order (kip·in²)	9.8168	9.8168	29.763	29.763

A.5 Critical load

Z8x2, $\lambda_1 = \pi/300 = 0.01047$ in⁻¹:

$$EC_w\lambda_1^4 = 0.002314 \text{ kip}, \quad GJ\lambda_1^2 = 0.001144 \text{ kip}, \tag{16}$$

$$q_{\text{cr}} = \frac{EC_w\lambda_1^4 + GJ\lambda_1^2}{a_y} = 0.0008709 \text{ kip/in} = 10.45 \text{ plf}. \tag{17}$$

Z8x2.5: $q_{\text{cr}} = 18.67$ plf.